ACSM's Guidelines for Exercise Testing and Prescription (11th Edition)

Authors: American College of Sports Medicine

Abstract & Overview

Source: ACSM Guidelines for Exercise Testing and Prescription (11th Edition) & MediaPipe Pose Architecture (Lugaresi et al., 2019)
Citation: American College of Sports Medicine. (2021). ACSM's guidelines for exercise testing and prescription. Wolters Kluwer.

1. Warm-Up Protocols and Exercise Safety

- General Warm-Up: 5 to 10 minutes of light dynamic cardiovascular activity (jogging in place, jumping jacks) to raise core body temperature and increase tissue elasticity.
- Dynamic Mobility: Focus on target joint mobility:
  - Ankle dorsiflexion and hip openers (world's greatest stretch, deep goblet squat holds) before squats and lunges.
  - Thoracic spine rotations, cat-camels, and band pull-aparts before shoulder presses and push-ups.
- Warm-Up Sets: Perform 1 to 2 sets with 40-50% target weight before working sets to groove motor pathways.

2. Fundamentals of 2D Vision-Based Joint Angle Measurement

- Vector Cosine Principle: Given joint coordinates A(x_1, y_1), B(x_2, y_2) (vertex), and C(x_3, y_3), the angle \theta at vertex B is computed via:
\vec{u} = A - B, \quad \vec{v} = C - B
\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}, \quad \theta = \arccos(\text{clamp}(\cos(\theta), -1.0, 1.0))
- Camera Perspective Invariance & Parallax: 2D angles represent projections of 3D kinematic movements onto the camera sensor plane. When movement occurs oblique to the sensor, projected angles may foreshorten. This underscores the necessity of the pre-detector Camera View Suitability Module to enforce appropriate viewpoint alignment.
- Hysteresis in Repetition Counting: Finite-state machines must incorporate hysteresis bands (e.g. 155° overhead threshold vs 95° bottom threshold) to prevent rapid state chatter or double counting around boundary transitions.